Modelling of Dynamical Systems

3D printer

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October 20, 2026

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1 Introduction

The mini-project involves the modelling of a 3D-printer in three different domains. The domains are Physical, Discrete, and Hybrid. In Figure 1 the schematic overview with the associate variable names and orientations are shown. The orientation and names of the corresponding physical quantity are respected in the derivation of the models. The mandatory variable names are given in Table 1. The formulas contain auxiliary variables which may not be given in this table.

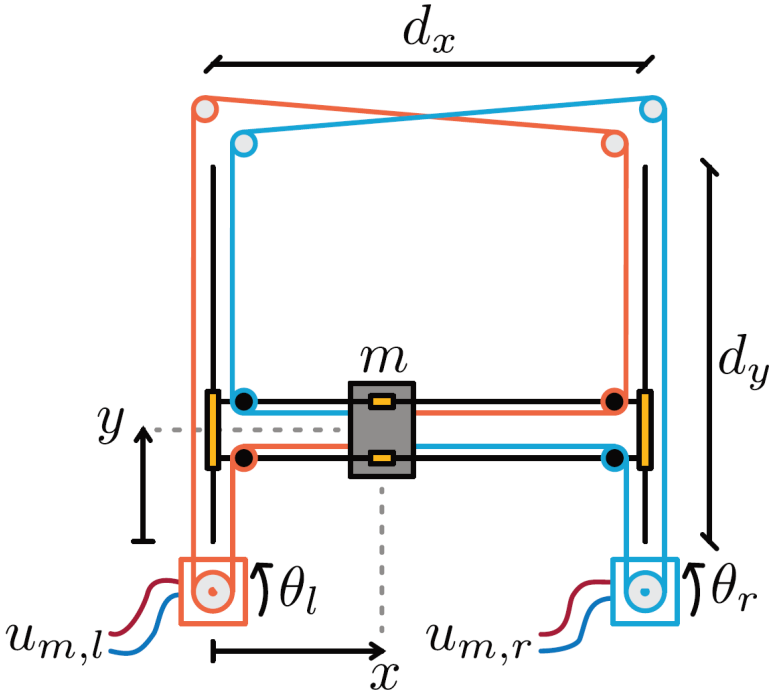


Figure 1: The schematic view of the 3D-printer.

Table 1: Variables and parameters of the 3D-printer.

Name	Symbol	Value	Unit
Motors and print head			
Motor angle left	θ_l		rad
Motor voltage left	$u_{m,l}$		V
Motor current left	$i_{m,l}$		A
Motor radius	r_m	11	mm
Motor inductance	L_m	10	mH
Motor resistance	R_m	90	Ω
Motor inertia	J_m	$8 \cdot 10^{-6}$	—
Motor constant	K_m	0.07	N m/A
Tool head position	x, y		m
Tool head mass	m	60	g
Nozzle and thermal parameters			
Nozzle voltage	u_n		V
Nozzle current	i_n		A
Nozzle capacitance	C_n		F
Nozzle thermal capacity	$C_{t,n}$		J/K
Nozzle thermal resistance	$R_{t,n}$	9	K/W
Nozzle resistance	R_n	8	Ω
Nozzle temperature	T_n	240	°C
Environmental temperature	T_e	300	K
Bearing friction			
Rolling bearing friction	μ_r	0.003	N s/m
Linear bearing friction, x-direction	$\mu_{l,x}$	0.002	N s/m
Linear bearing friction, y-direction	$\mu_{l,y}$	0.001	N s/m
Print bed and geometry			
Print bed temperature	T_b	105	°C
Print area dimension, x-direction	d_x	29	cm
Print area dimension, y-direction	d_y	29	cm

Right motor variables are analogous to those of the left motor. Empty value cells indicate variables or unspecified parameters. The unit of motor inertia is not specified in the assignment.

2 Physical Modelling

Question 1 Kinematics

The displacement of the belt by the driving motors having a radius of r_m can be expressed by the following equation

$$\Delta s = r_m \Delta\theta. \tag{1}$$

The leads to the equations for the left and right motor being described by

$$r_m \Delta\theta_l = \Delta x + \Delta y \tag{2}$$
$$r_m \Delta\theta_r = \Delta x - \Delta y. \tag{3}$$

In order to find Δx, one could add both equations

$$(\Delta x + \Delta y) + (\Delta x - \Delta y) = r_m \Delta\theta_l + r_m \Delta\theta_r, \tag{4}$$
$$2\Delta x = r_m \left(\Delta\theta_l + \Delta\theta_r\right), \tag{5}$$
$$\Delta x = \frac{r_m}{2} \left(\Delta\theta_l + \Delta\theta_r\right). \tag{6}$$

Similarly, the equation for Δy can be found by substraction

$$(\Delta x + \Delta y) - (\Delta x - \Delta y) = r_m \Delta\theta_l - r_m \Delta\theta_r, \tag{7}$$
$$2\Delta y = r_m \left(\Delta\theta_l - \Delta\theta_r\right), \tag{8}$$
$$\Delta y = \frac{r_m}{2} \left(\Delta\theta_l - \Delta\theta_r\right). \tag{9}$$

When the reference is taken to be zero (i.e., $x_0, y_0, \theta_{l,0}, \theta_{r,0} = 0$), the prefix Δ can be discarded. The earlier derived equations simplifies to

$$x = \frac{r_m}{2} \left(\theta_l + \theta_r\right), \tag{10}$$
$$y = \frac{r_m}{2} \left(\theta_l - \theta_r\right). \tag{11}$$

The velocities can be found by differentiation

$$\dot{x} = \frac{r_m}{2} \left(\dot{\theta}_l + \dot{\theta}_r\right), \tag{12}$$
$$\dot{y} = \frac{r_m}{2} \left(\dot{\theta}_l - \dot{\theta}_r\right). \tag{13}$$

The rotational speed is given by the derivative of the angle (i.e., $\dot{\theta} = \omega$). Therefore, the velocities can alternatively be denoted by

$$\dot{x} = \frac{r_m}{2} \left(\omega_l + \omega_r\right), \tag{14}$$
$$\dot{y} = \frac{r_m}{2} \left(\omega_l - \omega_r\right). \tag{15}$$

Question 2 Dynamics

The model of the DC motor coupled to a disk with an inertia is shown in Figure 2.

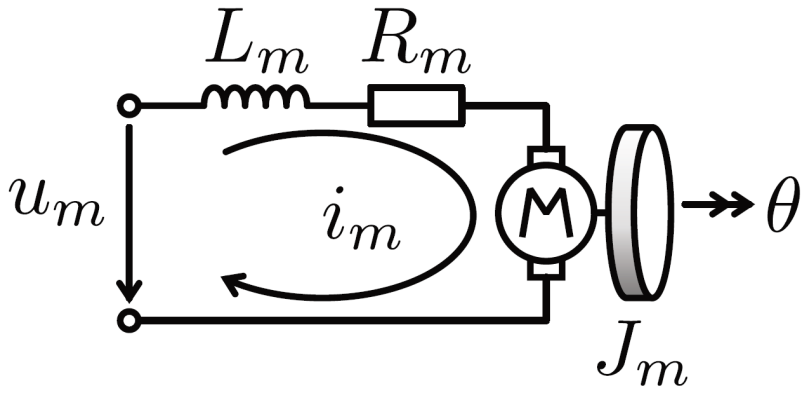


Figure 2: The proposed DC motor modelled as an LR circuit.

Applying Kirchhoff's voltage law on the schematic of Figure 2 leads to[1]

$$u_m = u_{L_m} + u_{R_m} + u_{\text{EMF}} \tag{16}$$

$$u_m = L_m \frac{di}{dt} + R_m i + K_m \dot{\theta} \tag{17}$$

$$\frac{di}{dt} = -\frac{R_m}{L_m} i - \frac{K_m}{L_m} \omega + \frac{1}{L_m} u_m \tag{18}$$

The rotation of the disc and its inertia is related to the current flowing through the motor.

$$\tau = K_m i \tag{19}$$

$$\tau = J \dot{\omega} \tag{20}$$

$$\dot{\omega} = \frac{K_m}{J} i. \tag{21}$$

The states are chosen as

$$\mathbf{x} = \left[\theta_l \, \theta_r \, \omega_l \, \omega_r \, i_{m,l} \, i_{m,r}\right]^\mathsf{T}. \tag{22}$$

The input vector is defined by

$$\mathbf{u} = \left[u_{m,l} \, u_{m,r}\right]^\mathsf{T}. \tag{23}$$

The positions of the 3D-printer are selected as the output vector of the system

$$\mathbf{y} = \left[x \, y\right]^\mathsf{T}. \tag{24}$$

[1]The derivative is normally denoted by a dot (e.g., $\dot{\theta}$) in this assignment except when the differential quantity is current (i.e., i). In that case, the explicit notation is used $\left(\text{i.e., } \frac{di}{dt}\right)$ in order to avoid confusion using a double dot (i.e., $\dot{i}$).

Therefore, leading to the elaborate state-space description

$$\dot{\mathbf{x}} = \underbrace{\begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{K_{m,l}}{J_l} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{K_{m,r}}{J_r} \\ 0 & 0 & -\frac{K_{m,l}}{L_{m,l}} & 0 & -\frac{R_{m,l}}{L_{m,l}} & 0 \\ 0 & 0 & 0 & -\frac{K_{m,r}}{L_{m,r}} & 0 & -\frac{R_{m,r}}{L_{m,r}} \end{bmatrix}}_{\mathbf{A}} \mathbf{x} + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{L_{m,l}} & 0 \\ 0 & \frac{1}{L_{m,r}} \end{bmatrix}}_{\mathbf{B}} \mathbf{u} \tag{25}$$

$$\mathbf{y} = \underbrace{\begin{bmatrix} \frac{r_m}{2} & \frac{r_m}{2} & 0 & 0 & 0 & 0 \\ \frac{r_m}{2} & -\frac{r_m}{2} & 0 & 0 & 0 & 0 \end{bmatrix}}_{\mathbf{C}} \mathbf{x}.$$

Question 3 **Decouple the x and y dynamics**

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